

Q.4) Given:-

$$h = 2\text{cm}$$

$$f = 10\text{cm}$$

$$u = -15\text{cm}$$

By applying mirror formula,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} + \left(-\frac{1}{15}\right) = \frac{1}{10}$$

$$\frac{1}{v} = \frac{1}{10} + \frac{1}{15}$$

$$\frac{1}{v} = \frac{3+2}{30}$$

$$\frac{1}{v} = \frac{5}{30}$$

$$v = \frac{30}{5}$$

$$\therefore v = 6\text{cm}$$

$$\therefore \text{Magnification} \Rightarrow -\frac{v}{u} = \frac{h'}{h}$$

$$= \frac{-6}{-15}$$

$$\Rightarrow \frac{-6}{-15} = \frac{h'}{2}$$

$$\Rightarrow 12 = 15h'$$

$$\therefore h' = \frac{12}{15}$$

The size of the image is very small and is formed between the pole and the focus on other side of the mirror.

Since we are using a convex mirror, so the image formed is virtual

5	10, 15
2	2, 3
	1, 1

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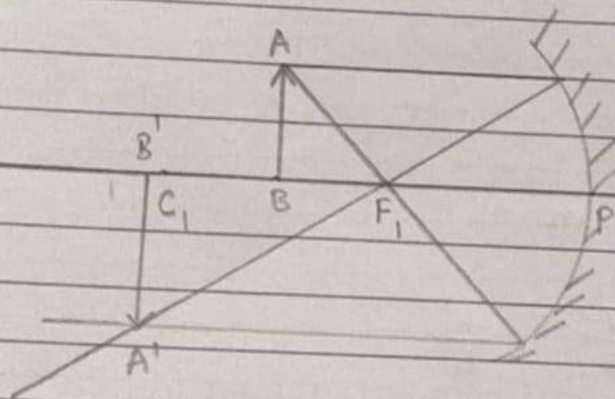
Physics

Q.1) Parallel beam of light

$$\begin{aligned} \text{Q.2) Focal Length} &= \frac{R}{2} \\ &= \frac{50}{2} \\ &= 25\text{cm} \end{aligned}$$

Since, the outer surface is made of aluminium, so the inner surface would be the reflecting surface. The mirror used will be concave mirror.

Q.3)



When object is kept between the centre of curvature and the principal focus of the concave mirror, a magnified image of the object is formed beyond the centre of curvature.